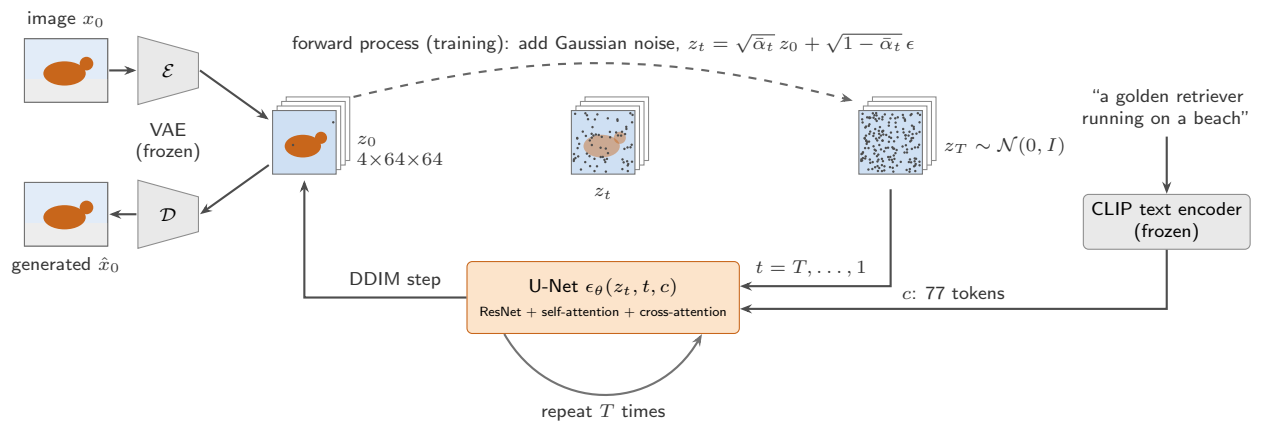

image x_0
$\mathcal{E}$
VAE
(frozen)
$\mathcal{D}$
generated $\hat{x}_0$
forward process (training): add Gaussian noise, $z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$
z_0
$4\times64\times64$
z_t
$z_T \sim \mathcal{N}(0, I)$
"a golden retriever running on a beach"
CLIP text encoder
(frozen)
DDIM step
U-Net $\epsilon_\theta(z_t, t, c)$
ResNet + self-attention + cross-attention
$t = T, \ldots, 1$
c: 77 tokens
repeat T times